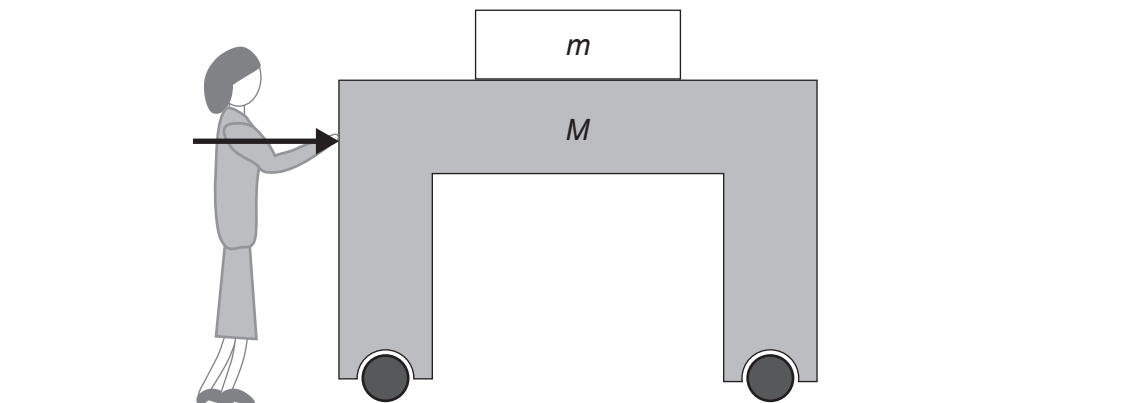


PAPER 1 / QUESTION 52

28 April 2022 / TZ2 / HL Paper 1 / original question 7 / 2222-6513

52. A book of mass m lies on top of a table of mass M that rolls freely along the ground. The coefficient of friction between the book and the table is μ . A person is pushing the rolling table.

What is the maximum acceleration of the table so that the book does not slide backwards relative to the table?



- A. $\frac{g}{\mu}$
B. μg
C. $\frac{mg}{M\mu}$
D. $\frac{m}{M} \mu g$